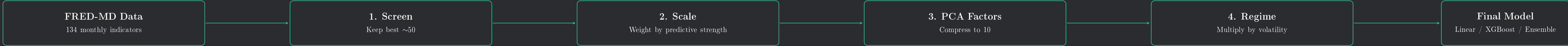


# Can Economic Data Predict the Stock Market?

## A Machine Learning Approach to S&P 500 Forecasting

### SSRF Pipeline: From Economic Data to Market Prediction



### What We Built

We built a pipeline that reads **130+ monthly economic statistics** (jobs, inflation, housing, interest rates) and predicts whether the S&P 500 will go up or down. We tested two horizons: next-month and next-quarter forecasts, simulating real-time trading over 26 years.

### Data

**Input:** 134 monthly economic indicators from the Federal Reserve's FRED-MD database (1979–2026). Covers employment, industrial production, housing, retail sales, interest rates, inflation, and consumer sentiment.  
**Target:** S&P 500 index returns from Yahoo Finance. Two targets: (1) next month's return, and (2) cumulative return over the next 3 months.  
**Size:** 540 months, 134 features. Training starts with 120 months (10 years) and expands walk-forward — no future data leaks in.

### Features

Raw indicators are noisy and highly correlated. We process them in three steps before feeding them to any model:

- Group-wise Screen** — Each indicator gets a t-statistic measuring how well it predicts returns on its own. Within each economic category, we keep only the ones that show some signal. For example, of 20 interest-rate indicators (3-month T-bill, 10-year yield, corporate spreads, etc.), only the strongest few survive. Same for jobs, inflation, housing — preventing any single category from flooding the model with noise. Typically 40–60 indicators remain.
- Predictive Scaling** — Each surviving indicator is multiplied by how strongly it has historically predicted returns. For example, if the unemployment rate has been a strong predictor, it gets amplified; if housing starts have been a weak one, it gets shrunk. This way the next step focuses on what actually matters for forecasting.
- Factor Extraction (PCA)** — The remaining indicators still move together (they all track the same economy). Principal Component Analysis compresses them into 10 uncorrelated “economic factors” — e.g., all employment indicators combine into a single labor factor, all interest rates into a rates factor, all inflation measures into a prices factor.

We also create **regime interaction terms** — multiplying each factor by recent market volatility — so the model adapts to calm vs. turbulent markets.

### Models

Seven models tested on the same pipeline:

| Model             | Description                          |
|-------------------|--------------------------------------|
| Elastic Net       | Linear model with feature selection  |
| Linear Regression | Simple linear combination of factors |
| XGBoost           | Gradient-boosted decision trees      |
| Random Forest     | Ensemble of many decision trees      |
| CatBoost          | Advanced tree model                  |
| MLP (Neural Net)  | 2-layer neural network               |
| Ensemble          | Average of Linear + XGBoost          |

**Baselines:** Always-long (predict market always goes up) and momentum (predict same direction as last month).

### Results

|                        | 1-Mo     | 3-Mo     | 3-Mo   |
|------------------------|----------|----------|--------|
| Model                  | Hit Rate | Hit Rate | Sharpe |
| CatBoost               | 56.9%    | 74.5%    | 0.85   |
| Ensemble               | 52.8%    | 70.3%    | 0.87   |
| XGBoost                | 56.7%    | 66.2%    | 0.82   |
| Linear                 | 50.7%    | 65.5%    | 0.74   |
| Random Forest          | 54.6%    | 62.8%    | 0.61   |
| Elastic Net            | 55.0%    | 59.3%    | 0.33   |
| MLP (Neural Net)       | 47.2%    | 62.1%    | 0.33   |
| Always-Long (baseline) | 62.5%    | 66.6%    | —      |
| Buy & Hold S&P 500     | —        | —        | 0.68   |

### Key Takeaways

- Monthly:** No model beats always-long (62.5%)
- 3-month:** CatBoost & Ensemble beat baseline
- No model predicts *how much* market moves
- Signal exists at quarterly horizon, not monthly

### Discussion

**Monthly:** All models fail. None beat the simple strategy of always predicting up (62.5% accuracy). This matches the well-known result that next-month stock returns are unpredictable from economic data.  
**Quarterly:** Real signal appears. CatBoost (74.5%) beats the 66.6% baseline by nearly 8 points. The Ensemble reaches 70.3% with the best risk-adjusted return (Sharpe 0.87).  
**But the signal is weak.** Models get *direction* right but can't predict *magnitude*. All have negative  $R^2_{OOS}$ , meaning a simple historical average predicts magnitudes better.  
**Leverage helps manage risk.** Asymmetric sizing (2.5× long, 0.25× short) cuts maximum loss from ~47% (buy & hold) to ~10%, though it slightly reduces Sharpe for the strongest models.

### What Didn't Work

- Extreme leverage** (25×): destroyed all portfolios
- Monthly returns:** hopeless at this horizon
- Magnitude prediction:** all  $R^2_{OOS}$  negative
- Transaction costs:** eat 1–2% of annual returns

### Future Work

- Model actual release lags of economic data (FRED reports come weeks after month-end)
- Replace volatility proxy with a Hidden Markov Model for better regime detection
- Full transaction cost modeling for production systems

### References

- Goyal & Welch (2008). A Comprehensive Look.
- Campbell & Thompson (2008). Predicting Excess Returns.
- McCracken & Ng (2016). FRED-MD Database.
- Zou & Hastie (2005). Elastic Net.

**Code:** [https://github.com/pellucide/sp500\\_macro\\_forecast](https://github.com/pellucide/sp500_macro_forecast)